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Data Science & Machine Learning

Assignment- 2

(10)

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1) How do you prune a decision Tree?

Answer

→ A decision tree can continue to grow indefinitely choosing splitting features and dividing into smaller & smaller partitions until each example is classified or the algorithm runs out of features to split on.

→ However, if the tree grows overly large, many of the decisions it makes will be overly specific and the model will have been overfitted to training data.

→ The process of pruning a decision tree involves reducing its size such that it generalizes better to unseen data.

→ One solution to this problem is to stop the tree from growing once it reaches a certain number of decisions or if the decision nodes contain only a small number of examples. This is called early stopping or pre-pruning the decision tree.

→ An alternative called post-pruning involves growing a tree that is too large then using pruning criteria based on the error rates at the node to reduce the size of the tree to a more appropriate level.

→ This is often a more effective approach than Pre-pruning because it is quite difficult to determine the optimal depth of a decision tree without gr

Classification rules represent knowledge in the form of if else statements that assign a class to unclassified examples.

They are specified in terms of an antecedent and a consequent, these form a hypothesis stating that "if this happens, then that happens".

The antecedent comprises certain combinations of feature values, while the consequent specifies the class value to assign if the rule's conditions are met.

Example

→ Antecedent

if cancel = "yes" then

~~Antecedent~~

return class = "bad" → Consequent

else

return class = "normal"

endif

remaining data

→ As rules are added, additional subsets of data are separated until the entire dataset has been covered and no more examples remain.

One rule Algorithm (1R - algorithm)

→ The way this algorithm works is simple for each feature, 1R divides the data into groups based on similar values of the feature

→ Then, for each segment, the algorithm predicts the majority class.

→ The error rate for the rule based on each feature is calculated, and the rule with the fewest errors is chosen as the one rule